Lecture 1: Introduction to Robotics & Recap of Linear Algebra

Iñigo Iturrate

Associate Professor

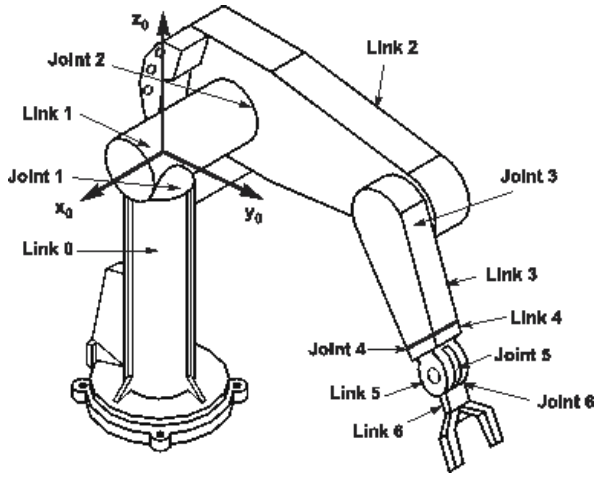
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What are we doing today?

1. **Introduction to Robotics & Recap of Linear Algebra and Mathematical Notation** ***(Today)***
2. Translations & Rotation Matrices
3. Other Representations for Orientation
4. Transformation Matrices
5. Forward Kinematics & DH Parameters
6. Schematic Representations of Robots, FK from DH, DH Parameters Practice
7. Inverse Kinematics
8. Kinematic Simulation
9. Angular Velocities, Velocity Propagation & Trajectory Generation
10. The Jacobian Matrix, Singularities
11. Analytical & Geometrical Jacobian, Numerical IK, Statics
12. Manipulability, More on the Robotic Systems Toolbox

Topics for Today

Part I:

- **What is a** ”**robot”**? What are (some) **different kinds of robots**?
- What **fields of study** are there within robotics?
  - **How does Robot Kinematics fit in** with this?
- Our first kinematic concept: **frames**
- **How is a robot defined** within Kinematics?
- What **mathematical tools** do we need **in order to use frames**?

Part II

- Recap of **Linear Algebra**: Vectors, Matrices, Spaces, Maps, Notation

Part III

- Getting started with **MATLAB**

Part I: Introduction to Robot Kinematics

What comes to mind when I say the word “Robot”?

What types of robots can you think of?

Some Types of Robots



Articulated Robots



SCARA
(Selective Compliance Assembly Robot Arm)



Delta Robots



Cartesian (XYZ)

This is what we will mostly work with



Mobile Robots
(Automated Guided Vehicles/ Autonomous Mobile Robots)



Drones
(Unmanned Aerial Vehicles)



Humanoids

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How are robots different from other automation?

Robots vs. Traditional Industrial Automation

Main difference: **flexibility.**

Traditional Industrial Automation
(e.g. CNC milling machine)



- Designed to **do one task** efficiently.
- Can be **programmed to perform variations** of a task.
- Typically, the **production line is designed around the specific machine**.

Robots
(e.g. Universal Robots UR3e)



- Designed for a **wide range of tasks**.
- Can be **programmed to perform many different tasks**.
- Typically, can be **integrated into an existing production line**.

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(Some) Classical Fields of Study Within Robotics

Robotics

Kinematics



Studies **movement** in the absence of forces and decribing relative locations of objects

Dynamics



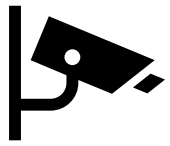
Studies **forces** and the **interaction** of robots **with their environment**

Control Theory



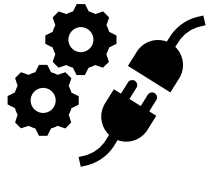
Studies **models** of inputs/outputs to **achieve a desired outcome**

Sensor Design



Studies the **construction and use of sensors** as input to a robotics system

Mechatronics Design



Studies the **construction of mechanisms** by combining **mechanics/ electronics**

. . .

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What are the assumptions of Kinematics?

1. As we said, *Kinematics* implies that:
   - We study **motion.**
   - We **ignore any forces**.

Discuss with your neighbors:
What does the lack of forces imply about the objects we study?

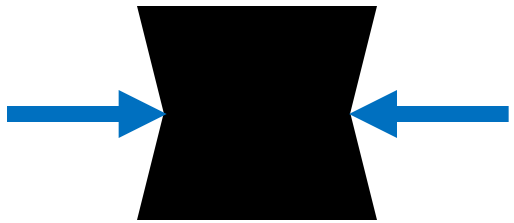
In other words, what property of objects could we know for sure holds true in a world where there are no forces?

What are the assumptions of Kinematics?

1. As we said, *Kinematics* implies that:
   - We study **motion.**
   - We **ignore any external forces**.

2. If we ignore external forces, this implies we are also **ignoring any deformations**.

3. If we ignore deformations, **all objects are considered rigid**.



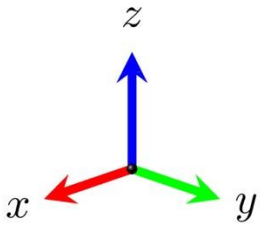


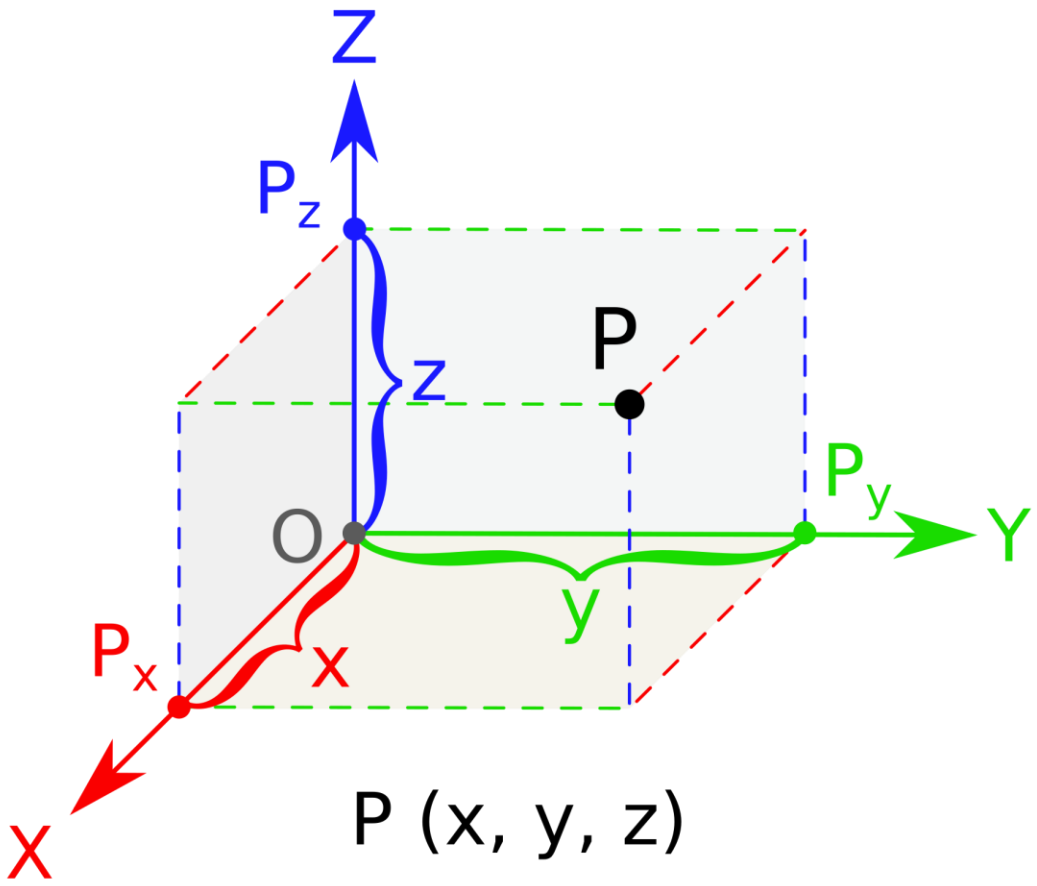




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Our First Kinematic Concept: The Frame





The **coordinate frame** is a concept you should know well:

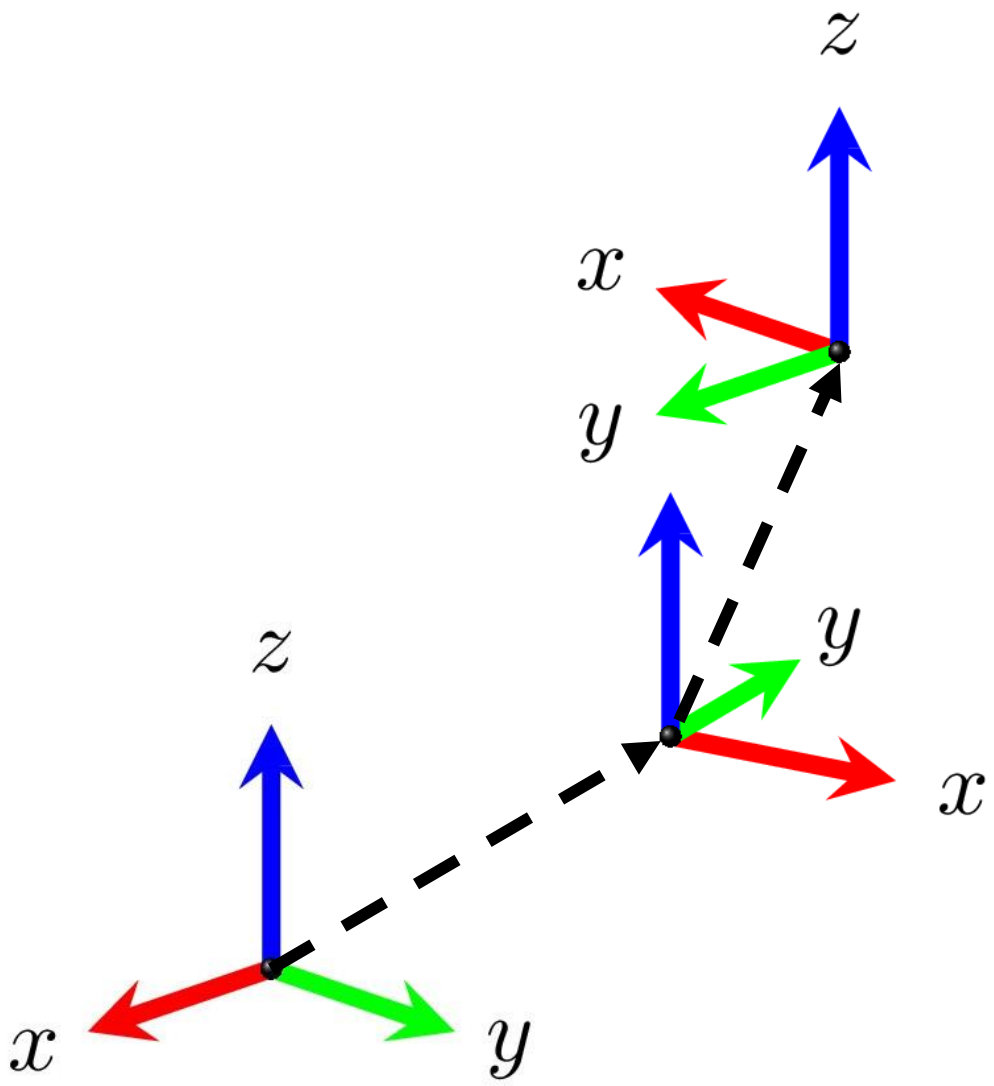
- Three (3) axes: *X, Y, Z*
- All axes are perpendicular to each other
- The location of a point, **P** is specified by three coordinates: *x, y, z*.

In high school:

- You have used it to draw graphs/figures in 3D.
- You have only considered **isolated static frames**.

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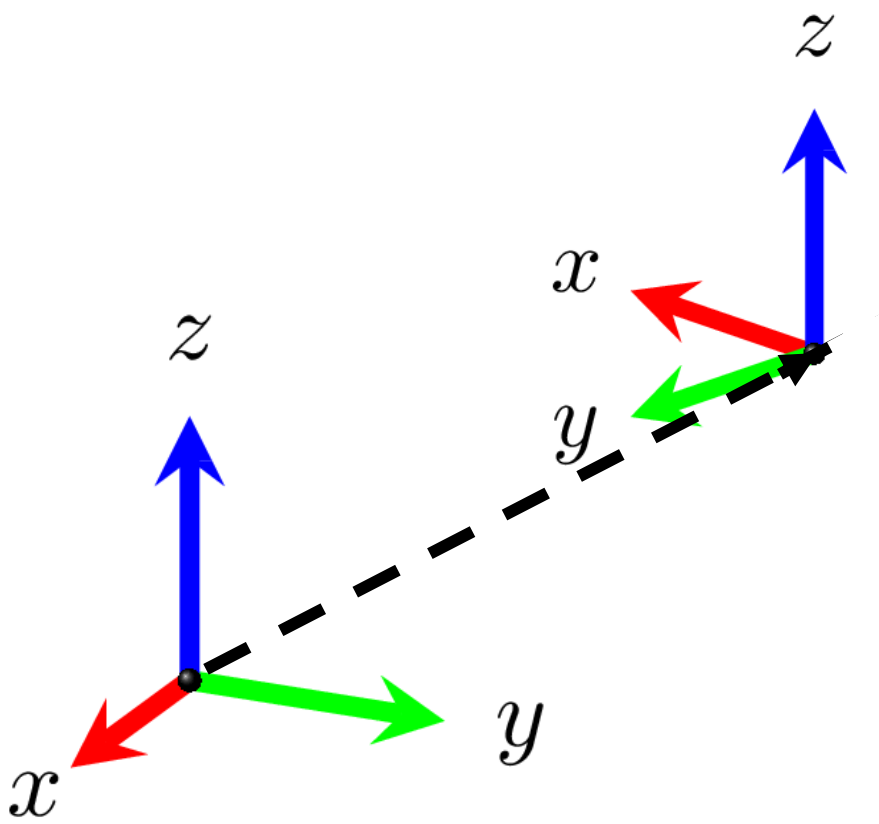
We can have more than one frame



- We can have **combinations of multiple frames**.
- We can **describe one frame**'s position and orientation **from another frame's point-of-view**.
- The two above concepts combined to form a **kinematic chain**.

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Our Frames Can Move



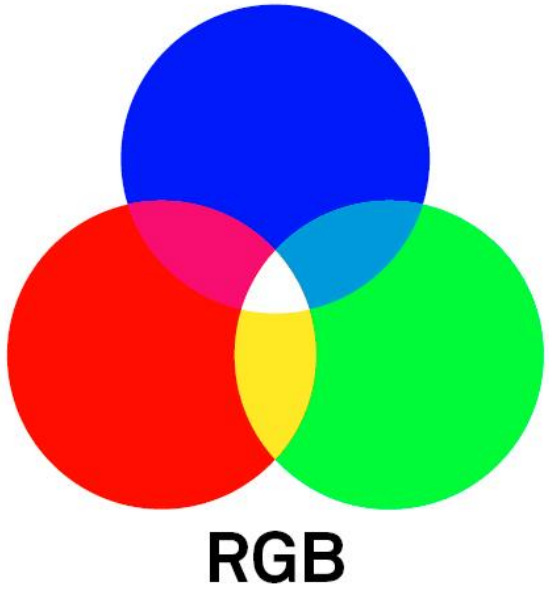
- Our **frames can move**.
- This motion can be **relative to other frames**.

The Right-Hand Rule

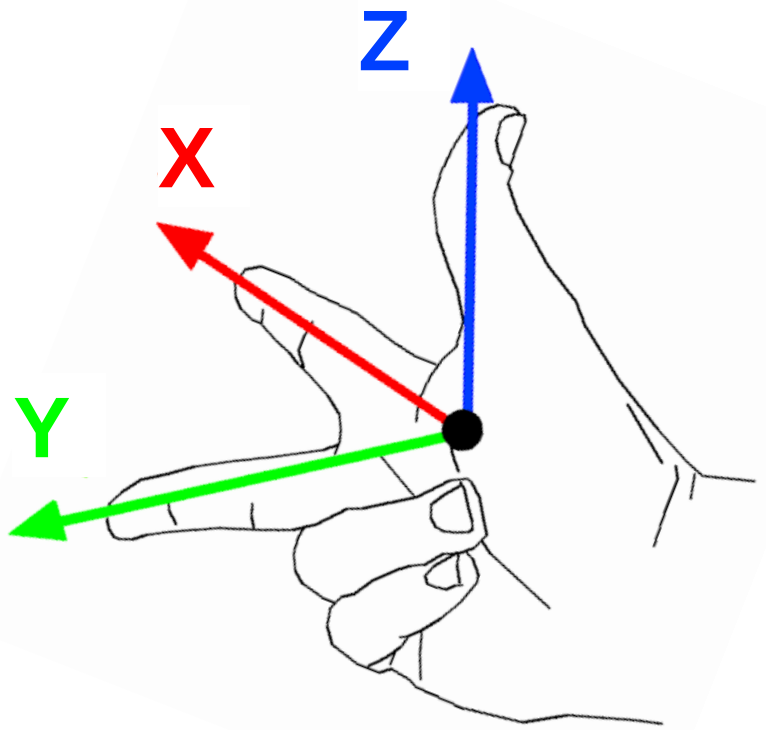
An easy way to remember which way the axes are pointing.

Remember three things:

1. The order of the colors given to the axes, *XYZ*, correspond to the usual order of colors:



2. Place your fingers like this:



3. It is called the **right-hand** rule. *Please,* use your right hand

(Sorry, lefties, apparently nobody thinks of you*).

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*Consider getting a job in graphics rendering engine development. Apparently they like you and often use left-handed coordinate systems.

So… what is a robot from the POV of Kinematics?

A robot is a series of:

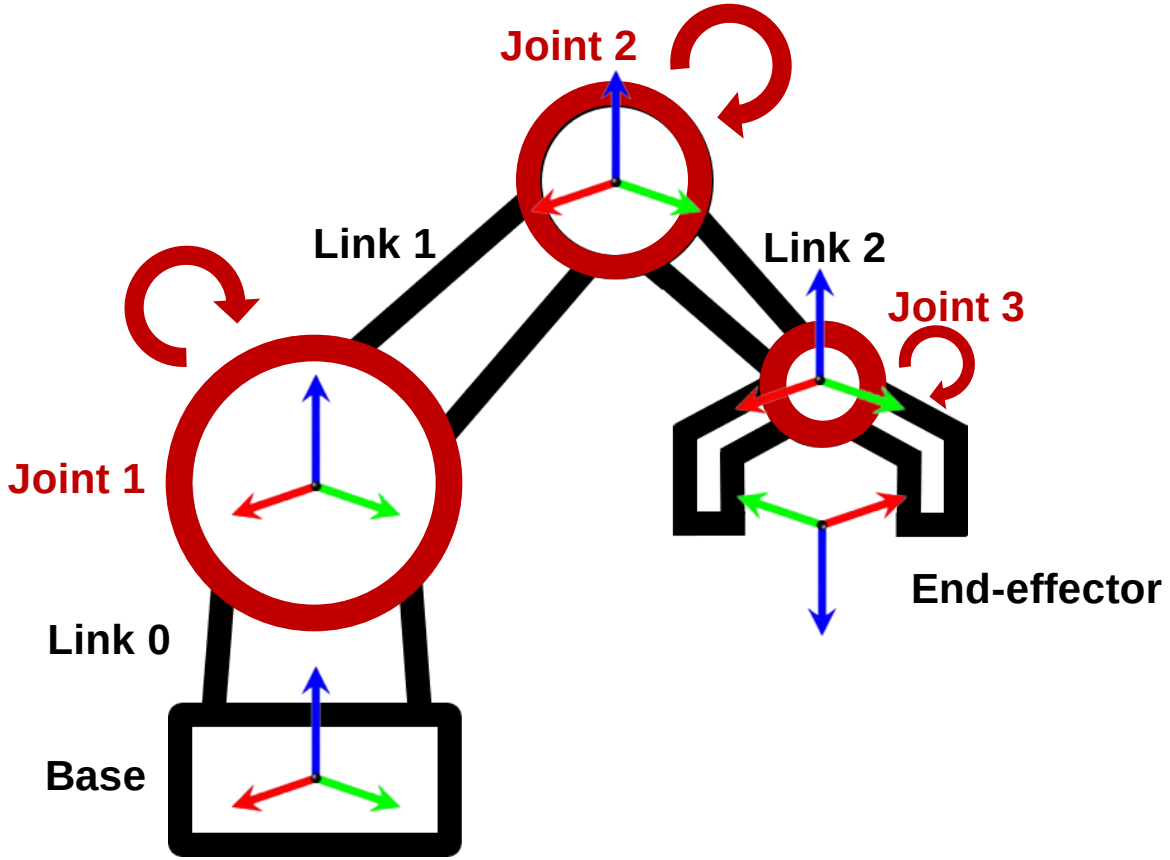
- Rigid **links**, which **do not move** on their own.
- **Joints**, which **allow relative movement** between links.

We can **place coordinate frames at each of the joints.**

This allows us to describe how:

- The **links move with respect to each other**.
- The **end-effector moves relative to the base**.

This representation is called a **Rigid Body Tree**.



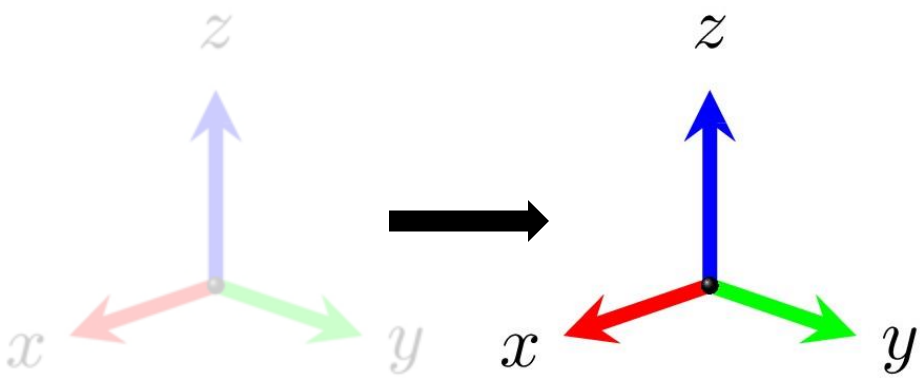
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What mathematical tools do we need?

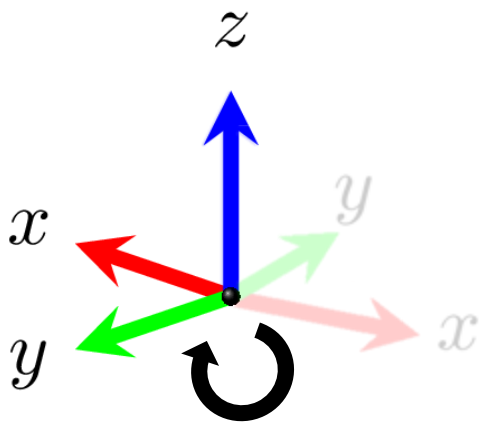
We now know (conceptually) that **we can use frames to describe the motion** of objects (or parts of a robot) relative to each other.

But how do we actually do this **in practice using math**? We need to be able to describe:

Difference in **position**
between frames (**translation**)



Difference in **orientation**
between frames (**rotation**)



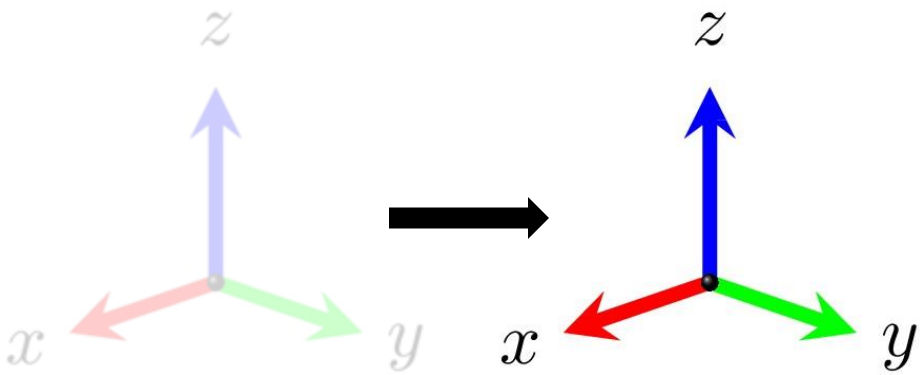
How can we do this?
Any ideas?

What mathematical tools do we need?

We now know (conceptually) that **we can use frames to describe the motion** of objects (or parts of a robot) relative to each other.

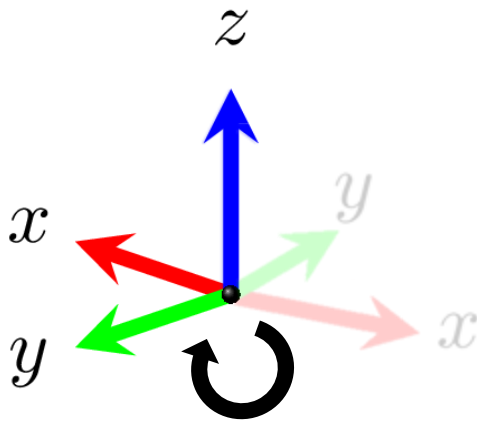
But how do we actually do this **in practice using math**? We need to be able to describe:

Difference in **position**
between frames (**translation**)



Vectors

Difference in **orientation**
between frames (**rotation**)



Vectors/Matrices
(For example. There are many ways)

Recap: What have we discussed so far?

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Recap: What have we discussed today

- Some different **kinds of robots**.
- Some different **fields of study within robotics** (including *Kinematics*, i.e. this course).
- The **key assumptions in Kinematics:**
  - **No forces**.
  - Every object is **rigid**.
- The concept of **coordinate frame**:
  - It can be **used to express locations** of e.g. points.
  - Multiple frames make up a **kinematic chain.**

Take home message:

A robot can be described as **rigid links** connected by **movable joints**, to which we can attach **coordinate frames** in order to describe their motion.

This representation is called a **rigid body tree**.

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Questions for the Next Lectures

- How do we **describe positions** within a frame?
- How do we **describe orientations** within a frame?
- How do we **combine positions and orientations**?

Part II: Recap of Linear Algebra

You: **Oh, no, here comes the math…**

Me: **Oh, yes, get used to it.**

Spaces

Fill in the following table:

Meaning	Symbol	Example
All integers		
All real numbers		
All positive real numbers		
All three-dimensional vectors		
All 3-by-3 matrices		
All orientations in 3D		
All points on a unit sphere		

Spaces

Meaning	Symbol	Example
All integers	$\mathbb{Z}$	1, -5, 239, …
All real numbers	$\mathbb{R}$	1.7, -0.3, …
All positive real numbers	$\mathbb{R}^+$	1.7, 39.01, 7, …
All three-dimensional vectors	$\mathbb{R}^3$	[3.00, -1.01, 2.74], …
All 3-by-3 matrices	$\mathbb{R}^{3\times3}$	$\begin{bmatrix} 1.2 & -0.1 & 0.0 \\ 3.5 & 1.0 & -9.7 \\ 1.6 & 6.5 & 0.1 \end{bmatrix}$, …
All orientations in 3D	$SO(3)$	*We will see this later*
All points on a unit sphere	S^2	[1, 0, 0], [0, 1, 0], …

Maps

A ***map*** is another name for a ***function*.**

We say a *function* is a *map* because it **connects** (”**maps**”) an input (in one space) to an output (in another space).

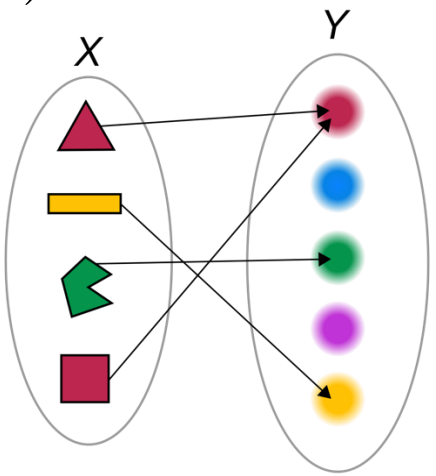
Examples:

$f(\mathbf{a}) : \mathbb{R}^3 \mapsto \mathbb{R}$ Maps from all real **3-dimensional vectors** to all **real numbers**

$f(\mathbf{a}) : \mathbb{R}^3 \mapsto \mathbb{R}^3$ Maps from all real **3-dimensional vectors** to all **3-dimensional vectors**

$f(a) : \mathbb{R} \mapsto \mathbb{R}^+$ Maps from all **real numbers** to all **positive real numbers**

$f(\mathbf{a}, \mathbf{b}) : \mathbb{R}^3 \times \mathbb{R}^3 \mapsto \mathbb{R}^3$ Maps from **two inputs: both 3-dimensional vectors** to all **3-dimensional vectors**



Maps from all **colored shapes** to all **colors**

Can you come up with example functions for each of the cases above?

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Vectors (I)

Row Vector: $\mathbf{v} = [v_1, v_2, v_3]$

Column Vector: $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

Point in Space: (column vector!) $\mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Length of a vector: $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{\sum_{i}^{N=3} v_i^2}$

$$\mathbf{v}, \mathbf{p} \in \mathbb{R}^3$$

Unit vector: $|v| = 1$

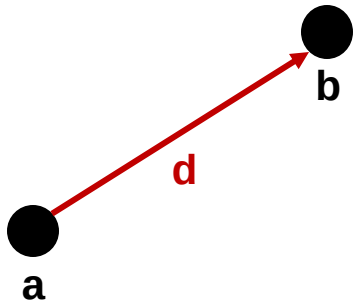
Normalization: (converting to unit length) $\hat{\mathbf{u}} = \dfrac{\mathbf{u}}{|\mathbf{u}|}$

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Vectors (II)

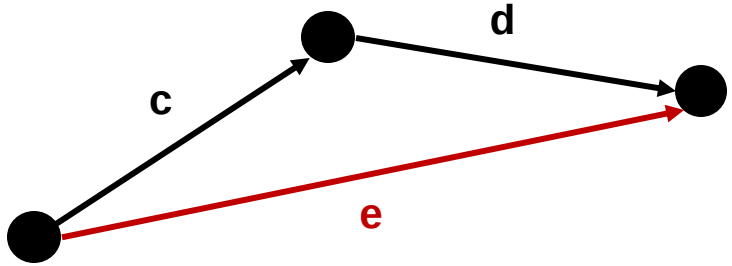
$$\mathbf{d} = \mathbf{b} - \mathbf{a} = \begin{bmatrix} b_1 - a_1 \\ b_2 - a_2 \\ b_3 - a_3 \end{bmatrix}$$

Vector Point Point



$$\mathbf{e} = \mathbf{c} + \mathbf{d} = \begin{bmatrix} c_1 + d_1 \\ c_2 + d_2 \\ c_3 + d_3 \end{bmatrix}$$

Vector Vector Vector



$$\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e} \in \mathbb{R}^3$$

Vectors (III): Dot/Scalar Product & Angle

Calculated as:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \sum_{i=1}^{N=3} a_n b_n = s$$

$\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ $s \in \mathbb{R}$

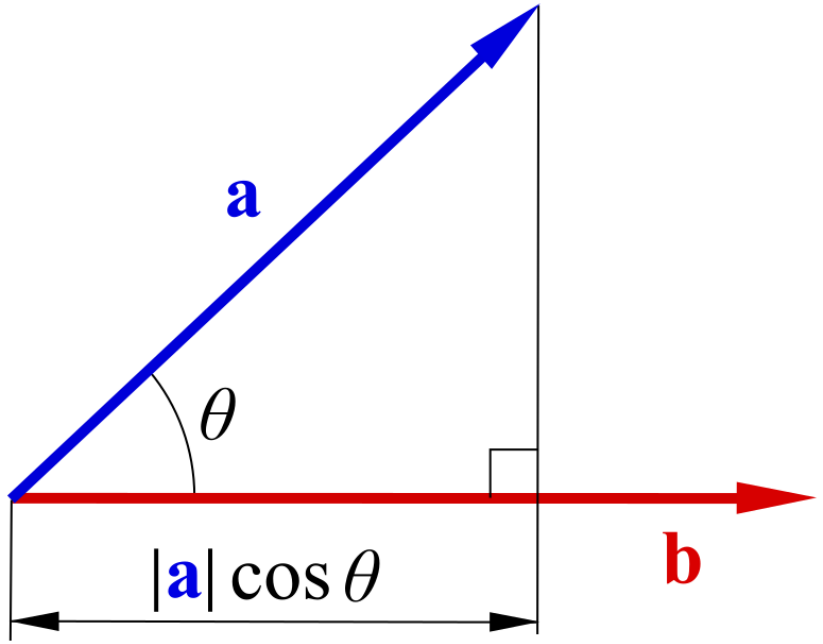
Important Properties:

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$ **Commutative!**

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(\theta)$$

Geometric Interpretation:

The dot product is related to the **projection** of *a* onto *b.* Specifically, this projection can be calculated as the dot product of *a* with *b normalized*.



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Vectors (IV): Cross/Vector Product

Calculated as:

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \mathbf{c}$$

$$\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$$

Important Properties:

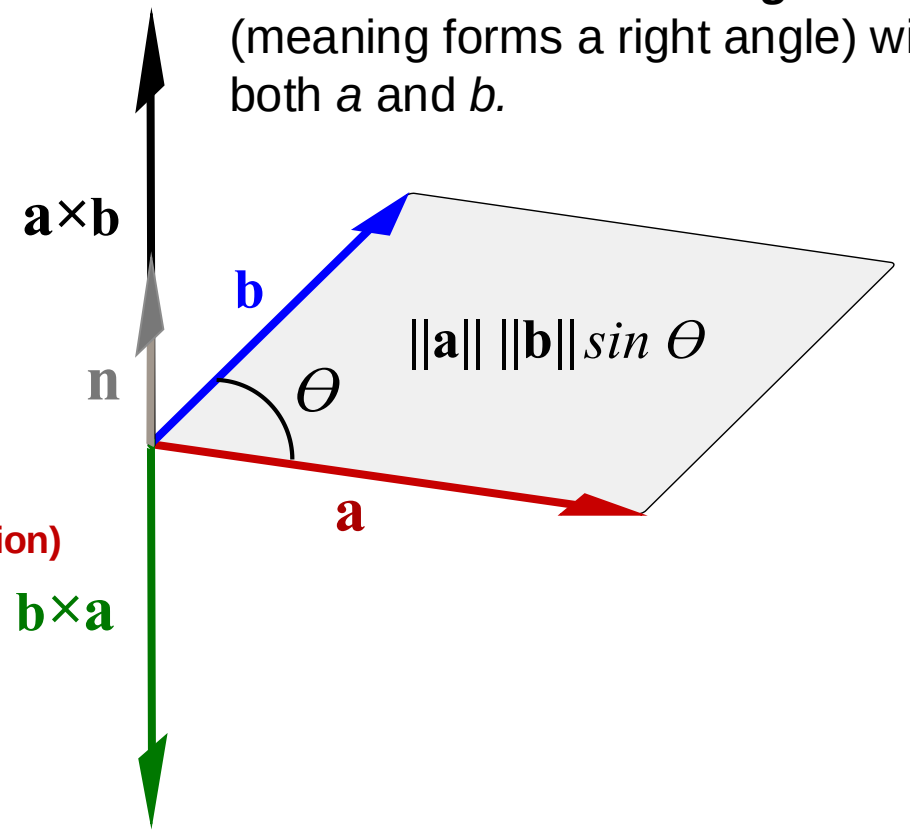
$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

Non Commutative!
(Same vector, opposite direction)

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin(\theta)\mathbf{n}$$

Geometric Interpretation:

The result of the cross product of *a, b* is a vector *c* that is ***orthogonal*** (meaning forms a right angle) with both *a* and *b.*



Orthogonality & Orthonormality

Orthogonal vectors are at a right-angle, i.e. 90°.

This implies that **the dot product of orthogonal vectors is 0.**

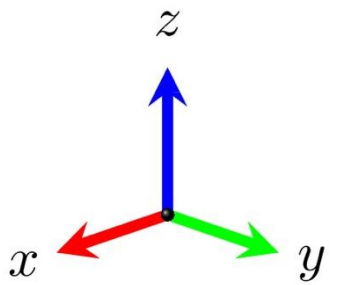
Orthonormality is fancy term for **orthogonal unit vectors**, i.e. vectors that fulfill the properties:

$$\mathbf{u} \cdot \mathbf{v} = 0$$ **Because they are *orthogonal***

$$|\mathbf{u}| = 1$$
$$|\mathbf{v}| = 1$$

Because they are *unit vectors*

I have already earlier in this lecture introduced a concept that depends on orthonormal vectors. Any idea what it is?



Maps (Revisited)

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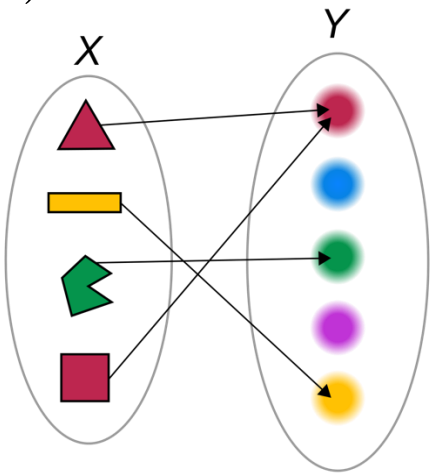
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$f(\mathbf{a}, \mathbf{b}) : \mathbb{R}^3 \times \mathbb{R}^3 \mapsto \mathbb{R}^3$ Maps from **two inputs: both 3-dimensional vectors** to all **3-dimensional vectors**



Maps from all **colored shapes** to all **colors**

Can you come up with example functions for each of the cases above?

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Matrices

Matrix:

m *rows*
n *columns*

$$\mathbf{A} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

$$\mathbf{A} \in \mathbb{R}^{m \times n}$$
$$m, n \in \mathbb{N}$$

Determinant (2x2): $\det(\mathbf{A}) = |\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}$

$$\mathbf{A} \in \mathbb{R}^{2 \times 2}$$

Determinant (3x3): $\det(\mathbf{A}) = |\mathbf{A}| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$$|\mathbf{A}| \in \mathbb{R}$$
$$\mathbf{A} \in \mathbb{R}^{3x3}$$

Visualize it as: $\det(\mathbf{A}) = |\mathbf{A}| = \overset{a_{11}}{\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}} - \overset{a_{12}}{\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}} + \overset{a_{13}}{\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}}$

Matrix Operations

Addition & Subtraction:

$$\mathbf{A} \pm \mathbf{B} = \begin{bmatrix} a_{11} \pm b_{11} & \dots & a_{1n} \pm b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & \dots & a_{mn} \pm b_{mn} \end{bmatrix} = \mathbf{C} = \begin{bmatrix} c_{11} & \dots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{m1} & \dots & c_{mn} \end{bmatrix}$$

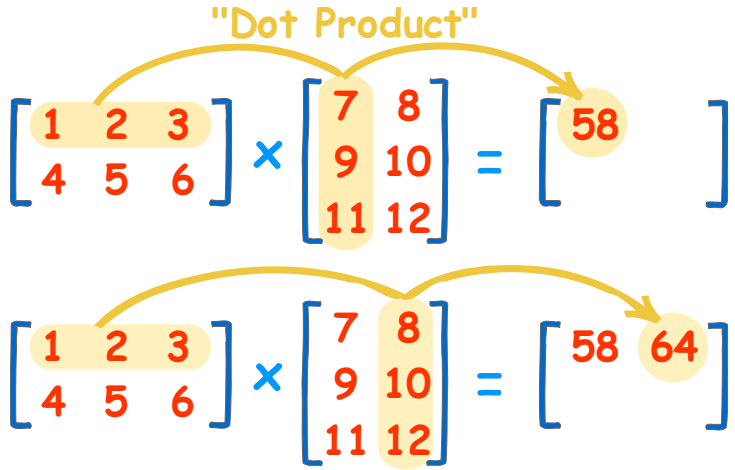
$$\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{m \times n}$$

Product:

$$\begin{bmatrix} \end{bmatrix}$$

$$\mathbf{A} \in \mathbb{R}^{m \times n}$$
$$\mathbf{B} \in \mathbb{R}^{n \times p}$$
$$\mathbf{C} \in \mathbb{R}^{m \times p}$$

Visualize it as:



Source: www.mathisfun.com

Inverse Matrix & Singular Matrix

A matrix, **A**, is invertible if there exists a matrix, **B**, such that: $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$

We then call **B** the **inverse** of **A**: $\mathbf{B} = \mathbf{A}^{-1}$

Only square matrices are invertible.

Inverse (2x2):

Inverse (3x3): Consider **A** as a matrix of three column vectors:

then:

Some Special Matrices

Singular Matrix:
(non-invertible)

$$\det(\mathbf{A}) = 0$$

Transposed Matrix:

$$\mathbf{A}^T, \quad \text{where} \quad a_{kj}^T = a_{jk}$$

(in other words, rows become columns)

Symmetric Matrix:

$$\mathbf{A} = \mathbf{A}^T$$

Skew-symmetric Matrix:

$$\mathbf{A}^T = -\mathbf{A}$$

Orthogonal Matrix:
(or Orthonormal)

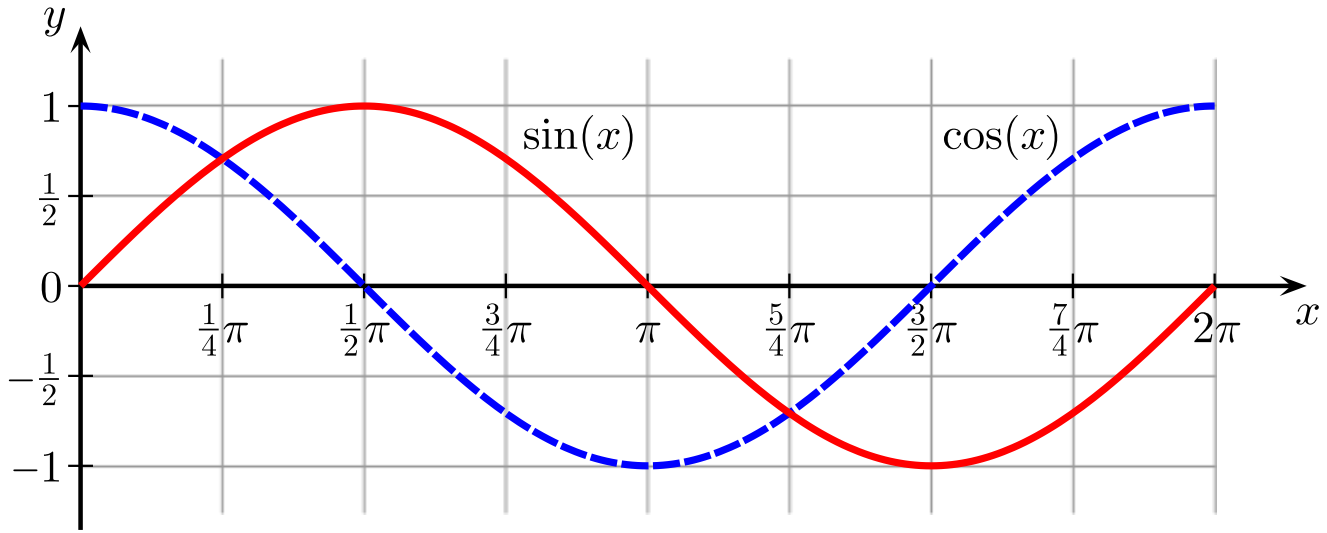
$$\mathbf{A}^T = \mathbf{A}^{-1}$$

$$\mathbf{A}^T\mathbf{A} = \mathbf{A}\mathbf{A}^T = \mathbf{I}_n$$

The **rows and columns** of an orthogonal matrix are **orthonormal vectors**.

Trigonometric Identities

Pythagorean Identity: $\sin^2 x + \cos^2 x = 1$



Reflections:

$$\sin(-x) = -\sin(x)$$
$$\cos(-x) = \cos(x)$$
$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x), \qquad \sin(\pi - x) = +\sin(x)$$
$$\cos\left(\frac{\pi}{2} - x\right) = \sin(x), \qquad \cos(\pi - x) = -\cos(x)$$

Shift:

$$\sin\left(x \pm \frac{\pi}{2}\right) = \pm\cos(x), \qquad \sin(x + \pi) = -\sin(x)$$
$$\cos\left(x \pm \frac{\pi}{2}\right) = \mp\sin(x), \qquad \cos(x + \pi) = -\cos(x)$$

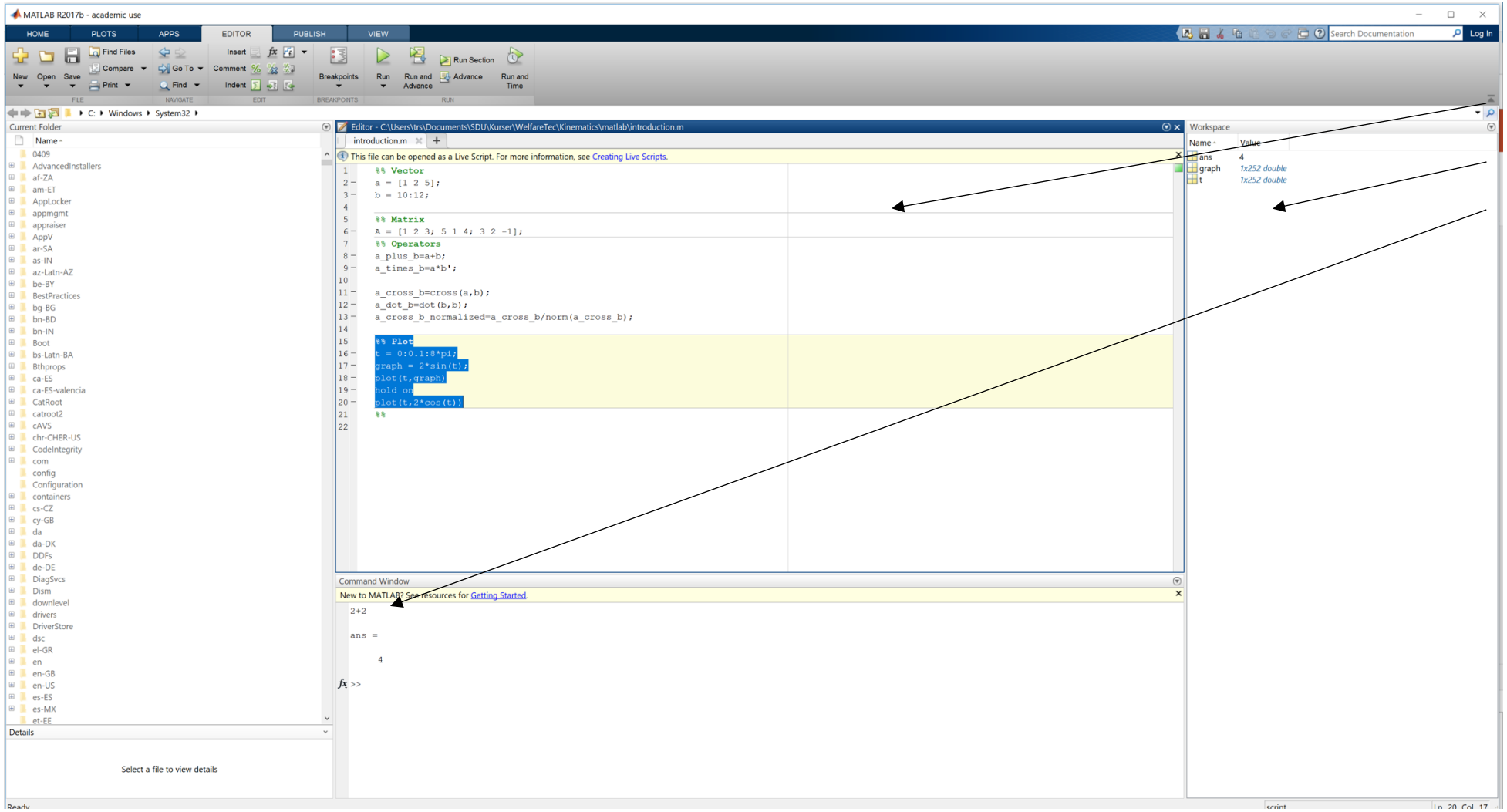
Addition:

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y)$$
$$\cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y)$$

SDU ROBOTICS

Part III: Introduction to MATLAB

Matlab: Getting Started



Editor

Workspace

Command Window

SDU ROBOTICS

Matlab: Vectors & Matrices

- vector

```
a = [1 2 5 1]

a =
     1   2   5   1
```

- matrix

```
x = [1 2 3; 5 1 4; 3 2 -1]

x =
     1     2     3
     5     1     4
     3     2    -1
```

- transpose

```
y = a.'    y =
                 1
                 2
                 5
                 1
```

Matlab: Ranges

- `t =1:10`

```
t =
     1     2     3     4     5     6     7     8     9    10
```

- `k =2:-0.5:-1`

```
k =
    2.0   1.5   1.0   0.5   0.0  -0.5  -1.0
```

- `x = [1:4; 5:8]`

```
x =
     1     2     3     4
     5     6     7     8
```

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Matlab: Arithmetic Operators

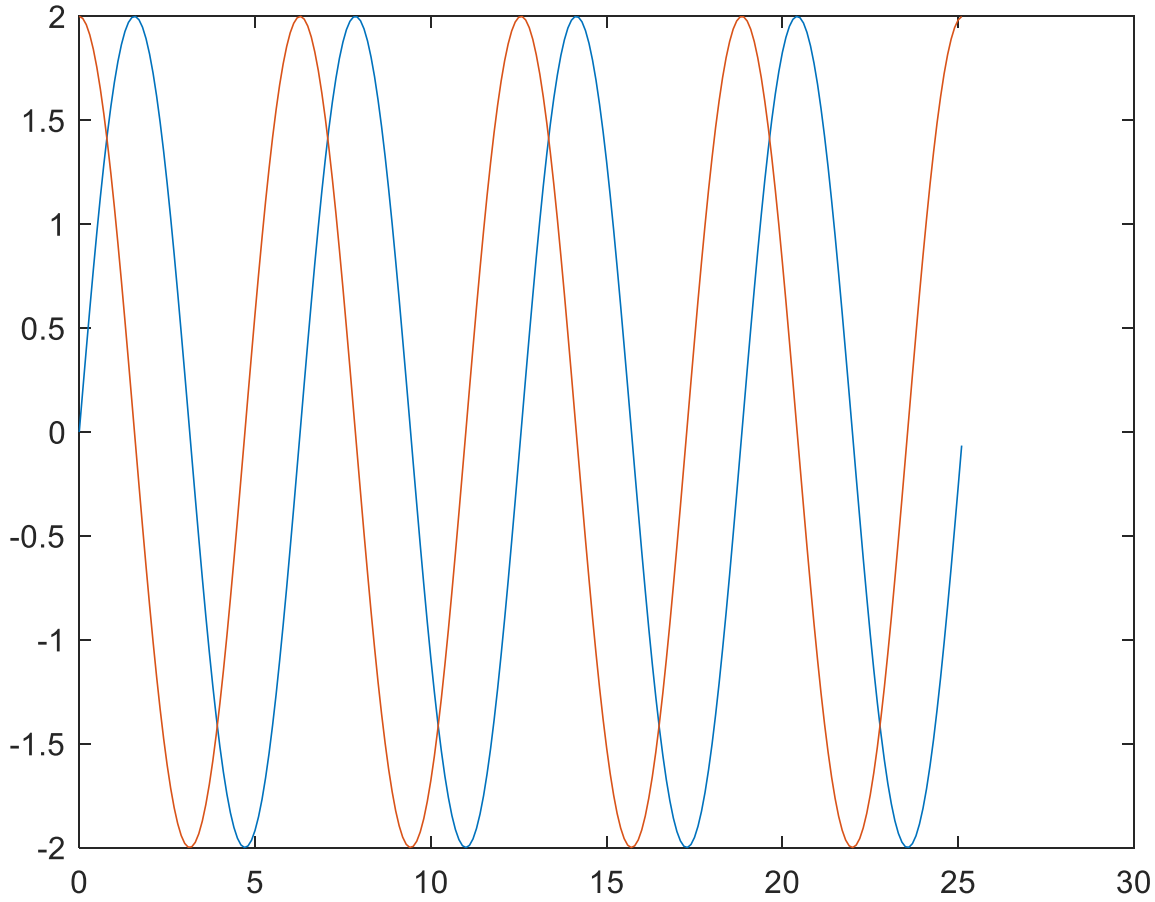
+ addition
- subtraction
* multiplication
/ division
^ power

Matlab: Some Built-in Functions

`cos(x)`	Cosine	`abs(x)`	Absolute value
`sin(x)`	Sine	`sign(x)`	Signum function
`tan(x)`	Tangent	`max(x)`	Maximum value
`acos(x)`	Arc cosine	`min(x)`	Minimum value
`asin(x)`	Arc sine	`ceil(x)`	Round towards $+\infty$
`atan(x)`	Arc tangent	`floor(x)`	Round towards $-\infty$
`exp(x)`	Exponential	`round(x)`	Round to nearest integer
`sqrt(x)`	Square root	`rem(x)`	Remainder after division
`log(x)`	Natural logarithm	`angle(x)`	Phase angle
`log10(x)`	Common logarithm	`conj(x)`	Complex conjugate

Matlab: Plotting

```
%% Plot
t = 0:0.1:8*pi;
graph = 2*sin(t);
plot(t,graph)
hold on
plot(t,2*cos(t))
```



Matlab: Generating Vectors from Functions

- zeros(M,N) MxN matrix of zeros

```
x = zeros(1,3)
x =
  0     0     0
```

- ones(M,N) MxN matrix of ones

```
x = ones(1,3)
x =
  1     1     1
```

- rand(M,N) MxN matrix of uniformly distributed random numbers on (0,1)

```
x = rand(1,3)
```

Matlab: Defining Functions

```
function [y1,...,yN] = myfun(x1,...,xM)

function [out] = addNumbers(x, y)
        out = x+y;
end


t = addNumbers(10,4)
t =
    14
```

Matlab: Defining Functions

- Chap. 1:
- 1.4
- 1.5
- 1.6
- 1.7

- Matlab:
- Implement your own function in Matlab that calculates the value of a polynomial of minimum second degree.

Thank you for today.

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